

$$(2) \quad E(MS_{trv}) = E\left(\frac{SS_{trv}}{a-1}\right) = E\left(\frac{1}{a-1} \left(\frac{1}{n} \sum_i y_{i.}^2 - \frac{y_{..}^2}{N}\right)\right)$$

We assume the effect model

$$y_{ij} = \mu + \tau_i + \epsilon_{ij} \quad \text{for } i, j \text{ in } a$$

$$= \frac{1}{a-1} E\left[\frac{1}{n} \sum_{i=1}^a \left(\sum_{j=1}^n \mu + \tau_i + \epsilon_{ij}\right)^2 - \frac{1}{N} \left(\sum_i \sum_j \mu + \tau_i + \epsilon_{ij}\right)^2\right]$$

$$\frac{1}{a-1} E\left[\frac{1}{n} \left(an\mu + n \sum_{j=1}^n \tau_i \right)\right]$$

$$\sum_i \tau_i = 0$$

$$= \frac{1}{a-1} E\left[\frac{1}{n} \sum_{i=1}^a \left(n\mu + n\tau_i + \sum_{j=1}^n \epsilon_{ij} \right)^2 - \frac{1}{N} \left(na\mu + n \sum_i \tau_i + \sum_i \sum_j \epsilon_{ij} \right)^2\right]$$

$$= \frac{1}{a-1} E\left[\frac{1}{n} \sum_i \left(n^2\mu^2 + n^2\tau_i^2 + \epsilon_{i.}^2 + 2n\mu\tau_i + 2n\mu\epsilon_{i.} + 2\tau_i\epsilon_{i.} \right) - \frac{1}{N} \left(N^2\mu^2 + \epsilon_{..}^2 + 2N\mu\epsilon_{..} \right)\right]$$

$$= \frac{1}{a-1} E\left[\frac{1}{n} \sum_i \left(n^2\mu^2 + n^2\tau_i^2 + \epsilon_{i.}^2 + 2n\mu\tau_i + 2n\mu\epsilon_{i.} + 2\tau_i\epsilon_{i.} \right) - \frac{1}{N} \left(N^2\mu^2 + \epsilon_{..}^2 + 2N\mu\epsilon_{..} \right)\right]$$

~~cancel~~

$$= \frac{1}{a-1} E\left[\frac{1}{n} \sum_i \left(n^2\mu^2 + n^2\tau_i^2 + \epsilon_{i.}^2 + 2n\mu\tau_i + 2n\mu\epsilon_{i.} + 2\tau_i\epsilon_{i.} \right) - \frac{1}{N} \left(N^2\mu^2 + \epsilon_{..}^2 + 2N\mu\epsilon_{..} \right)\right]$$

$$= \frac{1}{a-1} E\left[\left(n\mu^2 + n \sum_i \tau_i^2 + \frac{1}{n} \sum_i \epsilon_{i.}^2 + 2\mu\epsilon_{..} + 2 \sum_i \tau_i \epsilon_{i.} \right) - \left(N\mu^2 + 2\mu\epsilon_{..} + \frac{\epsilon_{..}^2}{N} \right)\right]$$

(Take expected value)

$$= \frac{a\sigma_\tau^2 + a\sigma^2 - \sigma^2}{a-1} = \frac{(a-1)\sigma^2 + (a-1)\sigma^2}{a-1} = \sigma^2 + n\sigma_\tau^2$$